

Roots (Part I) Homework

* Assume that all variables are positive.

Simplify.

① $\sqrt{28x^4y}$

② $\sqrt{36xy}$

③ $\sqrt{54x^5y^5}$

④ $\sqrt{90x^8y^6}$

⑤ $\sqrt{24x^{12}y^{13}}$

⑥ $-2\sqrt{72x^{15}y^4}$

⑦ $8x^6\sqrt{64x^4y^7}$

⑧ $2x^3\sqrt{8x^3} + 3\sqrt{18x^9}$

⑨ $-5y^2\sqrt{9y^2} + 7y^3 - 6y\sqrt{12y^7}$

⑩ $3x^4\sqrt{63x^2} - 5\sqrt{3x^3} - 5\sqrt{28x^{10}}$

⑪ $6y\sqrt{100y^{11}} + 9\sqrt{y^5} + 8y\sqrt{4y^3} - 7y^2\sqrt{y^9}$

$53y^6\sqrt{y} + 25y^2\sqrt{y}$

⑫ $\sqrt{5x} \cdot \sqrt{10x^3}$

⑬ $\sqrt{6x^7} \cdot \sqrt{8x^6}$

Answers

$2x^2\sqrt{7y}$

$6\sqrt{xy}$

$3x^2y^2\sqrt{6xy}$

$3x^4y^3\sqrt{10}$

$2x^6y^6\sqrt{6y}$

$-12x^7y^2\sqrt{2x}$

$64x^8y^3\sqrt{y}$

$13x^4\sqrt{2x}$

$-8y^3 - 12y^4\sqrt{3y}$

$-x^5\sqrt{7} - 5x\sqrt{3x}$

$5x^2\sqrt{2}$

$4x^6\sqrt{3x}$

⑭ $\sqrt{12x^5}(\sqrt{2x})$

⑮ $(3\sqrt{6x^{10}})(4\sqrt{3x^9})$

⑯ $\sqrt{5x^6} \cdot \sqrt{12x^8}$

⑰ $\sqrt{10x}(-2\sqrt{10x^2})$

Rationalize the denominators.

⑱ $\frac{4}{\sqrt{2}}$

$\frac{4\sqrt{2}}{2} = 2\sqrt{2}$

⑲ $\frac{8}{\sqrt{18}}$

$\frac{8\sqrt{2}}{6} = \frac{4\sqrt{2}}{3}$

⑳ $\sqrt{\frac{7}{81}}$

$\frac{\sqrt{7}}{9}$

㉑ $\sqrt{\frac{64}{11}}$

$\frac{8\sqrt{11}}{11}$

㉒ $\sqrt{\frac{3}{2}}$

$\frac{\sqrt{6}}{2}$

㉓ $\sqrt{\frac{24}{5}}$

$\frac{2\sqrt{30}}{5}$

$2x^3\sqrt{6}$

$36x^9\sqrt{2x}$

$2x^7\sqrt{15}$

$-20x\sqrt{x}$

$$\textcircled{24} \frac{1}{\sqrt{12}}$$

$$\textcircled{25} \sqrt{\frac{63}{40}}$$

$$\textcircled{26} \frac{\sqrt{8}}{\sqrt{18}}$$

$$\frac{\sqrt{3}}{6}$$

$$\frac{3\sqrt{70}}{20}$$

$$\frac{2}{3}$$